

Tagore Kosireddy

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EDUCATION

Michigan Technological University	Expected Aug 2028
PhD in Computer Science	Houghton, MI
Coursework: Computational Intelligence, Parallel Computing, Vision Language Models, Multi-Modal, NLP	
Michigan Technological University	Aug 2023 - April 2025
M.S. in Data Science, GPA: 3.94/4.0	Houghton, MI
Coursework: Machine Learning, Data Science, Advanced Data Mining, Predictive Modeling, Regression Analysis	
Gudlavaluru Engineering College	July 2018 - May 2022
B.Tech. in Electrical and Electronics Engineering, GPA: 8.7/10.0	Gudlavaluru, India

SKILLS

Languages:	Python, SQL, R, Java, C++, C
AI & ML & DL:	PyTorch, TensorFlow, Hugging Face, Transformers, Stable Baselines, Scikit-learn, Deep RL
Data Science:	Data Mining, Predictive Modeling, Statistics, Data Visualization (matplotlib, seaborn)
Other tools:	Git/GitHub, Flask, Linux, VsCode, Vim, NeoVim, Docker, Kubernetes, AWS Cloud, Azure

PROFESSIONAL EXPERIENCE

Michigan Technological University Graduate Research Assistant	Aug 2023 - Present
- Engineered and benchmarked autonomous RL system (PPO, DQN, A2C, SAC) in PyBullet for drone landing task - Created an LLM-based tutoring pipeline for seniors' digital competence; fine-tuned models (LoRA/QLoRA) - Developed financial language models integrated into AutoProphet, built LMForge for financial datasets - Automated staff processes via Python/Bash scripts and designed event materials for ICC/GLRC/MTU - Benchmarked Vision Language Models for autonomous vehicles in adverse winter weather conditions	
IHFC Tech Innovation Hub, IIT Delhi AI & Robotics Engineer	Jan 2023 - Aug 2023
- Designed ML models (PyTorch/TensorFlow) boosting robotic decision-making efficiency - Mentored 100+ students in RL, Python and ML for AI-driven projects - Guided 3 teams to win 1st prize in Regional National Robotics Competition	
Cognizant Technology Solutions Software Engineer	Jan 2022 - Dec 2022
- Engineered Java/Spring Boot APIs enhancing enterprise workflows, cutting latency through Hibernate - Optimized SQL queries (PostgreSQL) for high-volume data pipelines, reducing ingestion times by indexing/query - Built automated testing (JUnit/Mockito), slashing production bugs and boosting system reliability	
SmartInternz Artificial Intelligence Intern	Jan 2021 - Aug 2021
- Achieved 93% accuracy in PPE detection for warehouse safety using a ResNet model with transfer learning - Deployed a Python Flask application for real-time inference with a user-friendly HTML/CSS interface	

PUBLICATIONS

- PrivateNLP, ACL 2026 (Oral):** Loss Masking Under the Hood: Backdoor Concealment and Private Data Memorization in LLMs
- CustomNLP4U, ACL 2026 (Poster):** Small Language Models for the Democratization of Financial Literacy: Challenges and Opportunities
- WACV 2026 (LLVM-AD):** Benchmarking Vision-Language Models for Traffic Scene Understanding in Inclement Winter Weather: The AWDB Benchmark
- L2M2, ACL 2025:** Empirical Evaluation of Loss Masking to Selectively Prevent Memorization
- CustomNLP4U, EMNLP 2024:** Evaluation of small language models for financial literacy applications
- BabyLM Challenge, EMNLP CoNLL 2024:** Curriculum masking for LLM training on limited child language data

AWARDS

- First Place, NMU Programming Competition 2026** · Fourth Place, ICC Research Presentation Competition · Most Innovative Project, Winter Wonder Hackathon (MTU)

PROJECTS

- Lost it / Found it @ Tech:** MTU lost-and-found web app with posts, photo + time/location metadata, and embedding/vector search to auto-match items and notify users
- Free Parking @ Tech:** Parking helper showing permit zones + allowed hours; predicts spot availability from crowd updates + historical patterns (map UI + "can I park here now?" checker)